

武器系统动力学建模与仿真

多刚体系统的平面运动

发射动力学研究所

南京理工大学

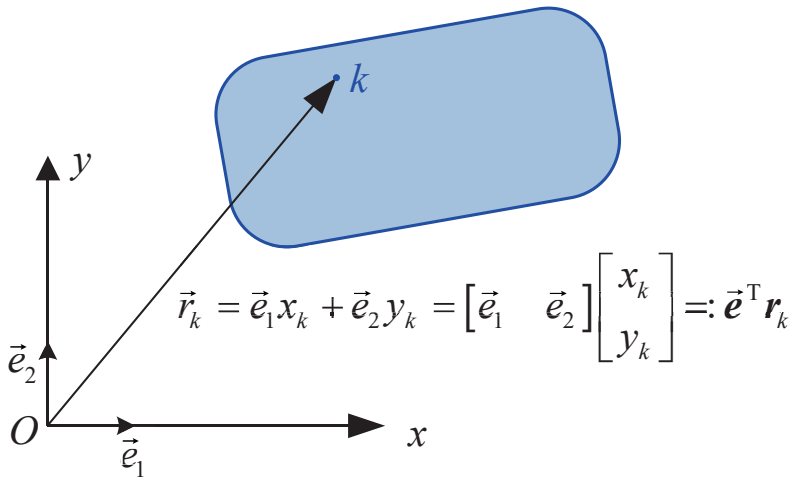
2025

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- ① 平面运动刚体运动学方程
- ② 平面运动刚体元件的动能
- ③ 平面运动刚体元件重力的虚功及其广义力
- ④ 弹性恢复力对应的广义力

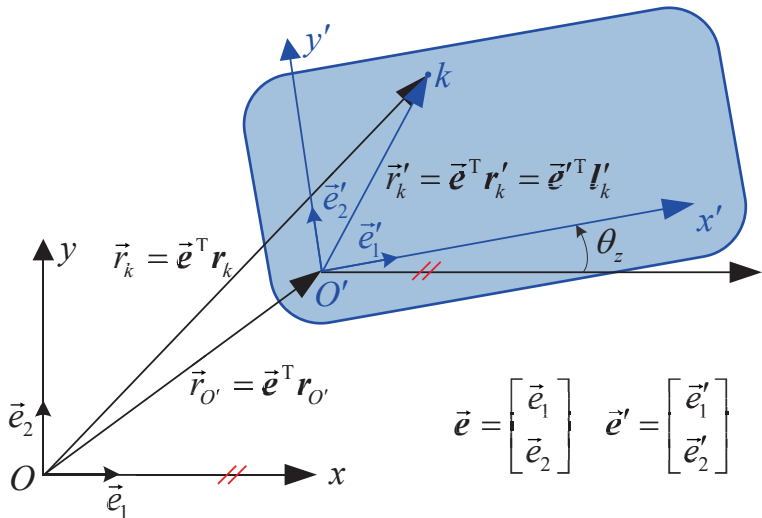
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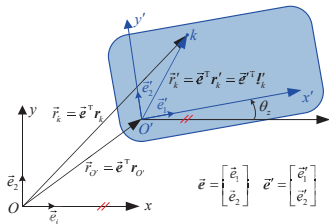


平面运动刚体元件的动能可表示为

$$T = \sum_{k=1}^N \frac{1}{2} m_k \dot{\mathbf{r}}_k^T \dot{\mathbf{r}}_k.$$



平面运动刚体元件的位置和姿态可由 $\mathbf{r}_{O'}$ 和 θ_z 表示；相应的，其平移速度和转动速度可由 $\dot{\mathbf{r}}_{O'}$ 和 $\dot{\theta}_z$ 表示。

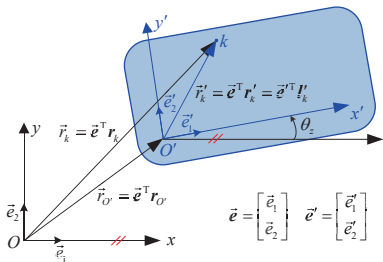


图中 $\mathbf{r}'_k$ 和 $\mathbf{l}'_k$ 分别为同一矢量 $\mathbf{r}'_k$ 在不同坐标系中的投影列阵，满足如下坐标变换式

$$\mathbf{r}'_k = \mathbf{A} \mathbf{l}'_k,$$

式中坐标变换矩阵 $\mathbf{A}$ 可用转角 θ_z 表示为

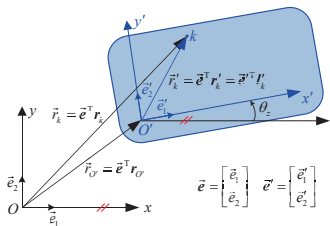
$$\mathbf{A} = \begin{bmatrix} \cos(\theta_z) & -\sin(\theta_z) \\ \sin(\theta_z) & \cos(\theta_z) \end{bmatrix}.$$



坐标变换式	$\mathbf{r}'_k = \mathbf{A} \mathbf{l}'_k$
坐标变换阵	$\mathbf{A} = \begin{bmatrix} \cos(\theta_z) & -\sin(\theta_z) \\ \sin(\theta_z) & \cos(\theta_z) \end{bmatrix}$

坐标变换矩阵 $\mathbf{A}$ 对时间的一阶导数满足

$$\begin{aligned}
 \dot{\mathbf{A}} &= \begin{bmatrix} -\sin(\theta_z)\dot{\theta}_z & -\cos(\theta_z)\dot{\theta}_z \\ \cos(\theta_z)\dot{\theta}_z & -\sin(\theta_z)\dot{\theta}_z \end{bmatrix} = \begin{bmatrix} -\sin(\theta_z) & -\cos(\theta_z) \\ \cos(\theta_z) & -\sin(\theta_z) \end{bmatrix} \dot{\theta}_z \\
 &= \begin{bmatrix} \cos(\theta_z) & -\sin(\theta_z) \\ \sin(\theta_z) & \cos(\theta_z) \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \dot{\theta}_z =: \mathbf{A} \mathbf{D} \dot{\theta}_z \\
 &= \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \cos(\theta_z) & -\sin(\theta_z) \\ \sin(\theta_z) & \cos(\theta_z) \end{bmatrix} \dot{\theta}_z =: \mathbf{D} \mathbf{A} \dot{\theta}_z
 \end{aligned}$$



微元 k 的位置坐标可表示为

坐标变换式	$\mathbf{r}'_k = \mathbf{A} \mathbf{l}'_k$
坐标变换阵	$\mathbf{A} = \begin{bmatrix} \cos(\theta_z) & -\sin(\theta_z) \\ \sin(\theta_z) & \cos(\theta_z) \end{bmatrix}$
矩阵导数	$\dot{\mathbf{A}} = \mathbf{A} \mathbf{D} \dot{\theta}_z, \mathbf{D} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

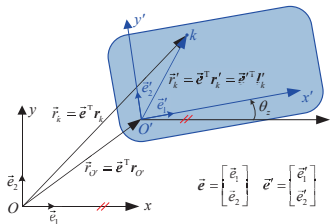
$$\begin{aligned} \mathbf{r}_k &= \mathbf{r}_{O'} + \mathbf{r}'_k \\ &= \mathbf{r}_{O'} + \mathbf{A} \mathbf{l}'_k, \end{aligned}$$

其对时间的一阶导数为

$$\dot{\mathbf{r}}_k = \dot{\mathbf{r}}_{O'} + \mathbf{A} \mathbf{D} \mathbf{l}'_k \dot{\theta}_z.$$

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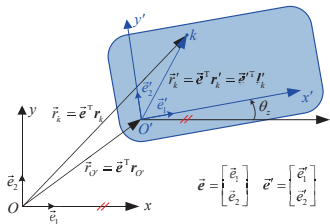
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坐标变换式	$\mathbf{r}'_k = \mathbf{A} \mathbf{l}'_k$
坐标变换阵	$\mathbf{A} = \begin{bmatrix} \cos(\theta_z) & -\sin(\theta_z) \\ \sin(\theta_z) & \cos(\theta_z) \end{bmatrix}$
矩阵导数	$\dot{\mathbf{A}} = \mathbf{A} \mathbf{D} \dot{\theta}_z, \quad \mathbf{D} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$
坐标导数	$\dot{\mathbf{r}}_k = \dot{\mathbf{r}}_{O'} + \mathbf{A} \mathbf{D} \mathbf{l}'_k \dot{\theta}_z$

平面运动刚体元件的动能可表示为

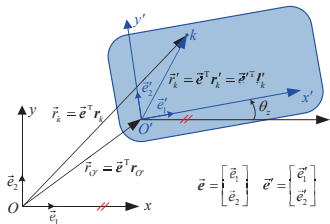
$$T = \sum_{k=1}^N \frac{1}{2} m_k \dot{\mathbf{r}}_k^T \dot{\mathbf{r}}_k$$



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平面运动刚体元件的动能可表示为

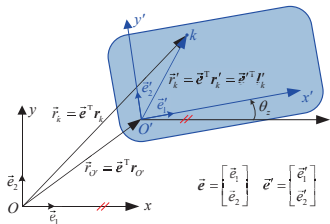
$$\begin{aligned}
 T &= \sum_{k=1}^N \frac{1}{2} m_k \dot{\mathbf{r}}_k^T \dot{\mathbf{r}}_k \\
 &= \sum_{k=1}^N \frac{1}{2} m_k \left(\dot{\mathbf{r}}_{O'} + \mathbf{A} \mathbf{D} \mathbf{l}'_k \dot{\theta}_z \right)^T \left(\dot{\mathbf{r}}_{O'} + \mathbf{A} \mathbf{D} \mathbf{l}'_k \dot{\theta}_z \right)
 \end{aligned}$$



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坐标变换阵	$\mathbf{A} = \begin{bmatrix} \cos(\theta_z) & -\sin(\theta_z) \\ \sin(\theta_z) & \cos(\theta_z) \end{bmatrix}$
矩阵导数	$\dot{\mathbf{A}} = \mathbf{A} \mathbf{D} \dot{\theta}_z, \mathbf{D} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$
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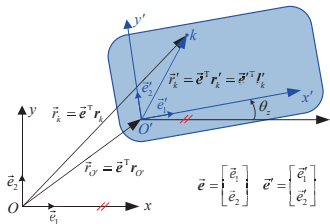
$$\begin{aligned}
 T &= \sum_{k=1}^N \frac{1}{2} m_k \left(\dot{\mathbf{r}}_{O'} + \mathbf{A} \mathbf{D} \mathbf{l}'_k \dot{\theta}_z \right)^T \left(\dot{\mathbf{r}}_{O'} + \mathbf{A} \mathbf{D} \mathbf{l}'_k \dot{\theta}_z \right) \\
 &= \sum_{k=1}^N \frac{1}{2} m_k \left(\begin{bmatrix} \mathbf{I}_2 & \mathbf{A} \mathbf{D} \mathbf{l}'_k \end{bmatrix} \begin{bmatrix} \dot{\mathbf{r}}_{O'} \\ \dot{\theta}_z \end{bmatrix} \right)^T \left(\begin{bmatrix} \mathbf{I}_2 & \mathbf{A} \mathbf{D} \mathbf{l}'_k \end{bmatrix} \begin{bmatrix} \dot{\mathbf{r}}_{O'} \\ \dot{\theta}_z \end{bmatrix} \right)
 \end{aligned}$$



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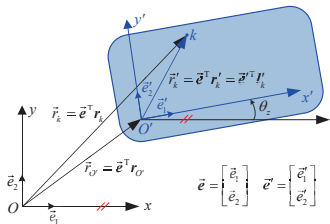
$$\begin{aligned}
 T &= \sum_{k=1}^N \frac{1}{2} m_k \left(\begin{bmatrix} \mathbf{I}_2 & \mathbf{A} \mathbf{D} \mathbf{l}'_k \end{bmatrix} \begin{bmatrix} \dot{\mathbf{r}}_{O'} \\ \dot{\theta}_z \end{bmatrix} \right)^T \left(\begin{bmatrix} \mathbf{I}_2 & \mathbf{A} \mathbf{D} \mathbf{l}'_k \end{bmatrix} \begin{bmatrix} \dot{\mathbf{r}}_{O'} \\ \dot{\theta}_z \end{bmatrix} \right) \\
 &= \sum_{k=1}^N \frac{1}{2} m_k \begin{bmatrix} \dot{\mathbf{r}}_{O'} \\ \dot{\theta}_z \end{bmatrix}^T \begin{bmatrix} \mathbf{I}_2 & \mathbf{A} \mathbf{D} \mathbf{l}'_k \end{bmatrix}^T \begin{bmatrix} \mathbf{I}_2 & \mathbf{A} \mathbf{D} \mathbf{l}'_k \end{bmatrix} \begin{bmatrix} \dot{\mathbf{r}}_{O'} \\ \dot{\theta}_z \end{bmatrix}
 \end{aligned}$$



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平面运动刚体元件的动能可表示为

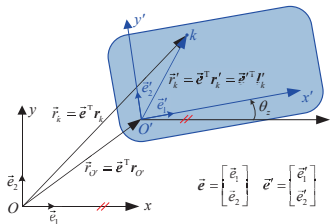
$$\begin{aligned}
 T &= \sum_{k=1}^N \frac{1}{2} m_k \begin{bmatrix} \dot{\mathbf{r}}_{O'} \\ \dot{\theta}_z \end{bmatrix}^T \begin{bmatrix} \mathbf{I}_2 & \mathbf{A} \mathbf{D} \mathbf{l}'_k \end{bmatrix}^T \begin{bmatrix} \mathbf{I}_2 & \mathbf{A} \mathbf{D} \mathbf{l}'_k \end{bmatrix} \begin{bmatrix} \dot{\mathbf{r}}_{O'} \\ \dot{\theta}_z \end{bmatrix} \\
 &= \sum_{k=1}^N \frac{1}{2} m_k \begin{bmatrix} \dot{\mathbf{r}}_{O'}^T & \dot{\theta}_z \end{bmatrix} \begin{bmatrix} \mathbf{I}_2 \\ \mathbf{l}'_k{}^T \mathbf{D}^T \mathbf{A}^T \end{bmatrix} \begin{bmatrix} \mathbf{I}_2 & \mathbf{A} \mathbf{D} \mathbf{l}'_k \end{bmatrix} \begin{bmatrix} \dot{\mathbf{r}}_{O'} \\ \dot{\theta}_z \end{bmatrix}
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$$\begin{aligned}
 T &= \sum_{k=1}^N \frac{1}{2} m_k \begin{bmatrix} \dot{\mathbf{r}}_{O'}^T & \dot{\theta}_z \end{bmatrix} \begin{bmatrix} \mathbf{I}_2 \\ \mathbf{l}'_k{}^T \mathbf{D}^T \mathbf{A}^T \end{bmatrix} \begin{bmatrix} \mathbf{I}_2 & \mathbf{A} \mathbf{D} \mathbf{l}'_k \end{bmatrix} \begin{bmatrix} \dot{\mathbf{r}}_{O'} \\ \dot{\theta}_z \end{bmatrix} \\
 &= \sum_{k=1}^N \frac{1}{2} m_k \begin{bmatrix} \dot{\mathbf{r}}_{O'}^T & \dot{\theta}_z \end{bmatrix} \begin{bmatrix} \mathbf{I}_2 & \mathbf{A} \mathbf{D} \mathbf{l}'_k \\ \mathbf{l}'_k{}^T \mathbf{D}^T \mathbf{A}^T & \mathbf{l}'_k{}^T \mathbf{D}^T \mathbf{A}^T \mathbf{A} \mathbf{D} \mathbf{l}'_k \end{bmatrix} \begin{bmatrix} \dot{\mathbf{r}}_{O'} \\ \dot{\theta}_z \end{bmatrix}
 \end{aligned}$$



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平面运动刚体元件的动能可表示为

$$\begin{aligned}
 T &= \sum_{k=1}^N \frac{1}{2} m_k \begin{bmatrix} \dot{\mathbf{r}}_{O'}^T & \dot{\theta}_z \end{bmatrix} \begin{bmatrix} \mathbf{I}_2 & \mathbf{A} \mathbf{D} \mathbf{l}'_k \\ \mathbf{l}'_k{}^T \mathbf{D}^T \mathbf{A}^T & \mathbf{l}'_k{}^T \mathbf{D}^T \mathbf{A}^T \mathbf{A} \mathbf{D} \mathbf{l}'_k \end{bmatrix} \begin{bmatrix} \dot{\mathbf{r}}_{O'} \\ \dot{\theta}_z \end{bmatrix} \\
 &= \frac{1}{2} \begin{bmatrix} \dot{\mathbf{r}}_{O'}^T & \dot{\theta}_z \end{bmatrix} \left(\sum_{k=1}^N m_k \begin{bmatrix} \mathbf{I}_2 & \mathbf{A} \mathbf{D} \mathbf{l}'_k \\ \mathbf{l}'_k{}^T \mathbf{D}^T \mathbf{A}^T & \mathbf{l}'_k{}^T \mathbf{l}'_k \end{bmatrix} \right) \begin{bmatrix} \dot{\mathbf{r}}_{O'} \\ \dot{\theta}_z \end{bmatrix} \\
 &=: \frac{1}{2} \begin{bmatrix} \dot{\mathbf{r}}_{O'}^T & \dot{\theta}_z \end{bmatrix} \mathbf{M} \begin{bmatrix} \dot{\mathbf{r}}_{O'} \\ \dot{\theta}_z \end{bmatrix}.
 \end{aligned}$$

平面运动刚体元件的质量矩阵

$$\begin{aligned}
 \mathbf{M} &= \sum_{k=1}^N m_k \begin{bmatrix} \mathbf{I}_2 & \mathbf{A} \mathbf{D} \mathbf{l}'_k \\ \mathbf{l}'_k{}^T \mathbf{D}^T \mathbf{A}^T & \mathbf{l}'_k{}^T \mathbf{l}'_k \end{bmatrix} \\
 &= \begin{bmatrix} \sum_{k=1}^N m_k \mathbf{I}_2 & \sum_{k=1}^N m_k \mathbf{A} \mathbf{D} \mathbf{l}'_k \\ \sum_{k=1}^N m_k \mathbf{l}'_k{}^T \mathbf{D}^T \mathbf{A}^T & \sum_{k=1}^N m_k \mathbf{l}'_k{}^T \mathbf{l}'_k \end{bmatrix} \\
 &= \begin{bmatrix} \sum_{k=1}^N m_k \mathbf{I}_2 & \mathbf{A} \mathbf{D} \sum_{k=1}^N m_k \mathbf{l}'_k \\ \left(\sum_{k=1}^N m_k \mathbf{l}'_k \right)^T \mathbf{D}^T \mathbf{A}^T & \sum_{k=1}^N m_k \mathbf{l}'_k{}^T \mathbf{l}'_k \end{bmatrix} \\
 &=: \begin{bmatrix} m \mathbf{I}_2 & \mathbf{A} \mathbf{D} m \mathbf{l}'_C \\ m \mathbf{l}'_C{}^T \mathbf{D}^T \mathbf{A}^T & J_{z,O'} \end{bmatrix}
 \end{aligned}$$

式中

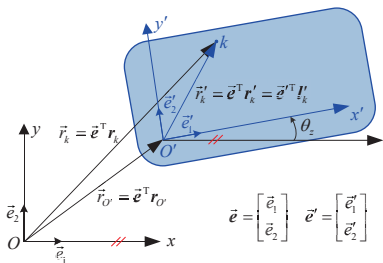
$$m = \sum_{k=1}^N m_k, \quad \mathbf{l}'_C = \frac{\sum_{k=1}^N m_k \mathbf{l}'_k}{\sum_{k=1}^N m_k}, \quad J_{z,O'} = \sum_{k=1}^N m_k \mathbf{l}'_k{}^T \mathbf{l}'_k.$$

平面运动刚体元件的动能

$$\begin{aligned} T &= \sum_{k=1}^N \frac{1}{2} m_k \dot{\mathbf{r}}_k^T \dot{\mathbf{r}}_k \\ &= \frac{1}{2} \begin{bmatrix} \dot{\mathbf{r}}_{O'}^T & \dot{\theta}_z \end{bmatrix} \begin{bmatrix} m\mathbf{I}_2 & \mathbf{A}\mathbf{D}m\mathbf{l}'_C \\ m\mathbf{l}'_C{}^T \mathbf{D}^T \mathbf{A}^T & J_{z,O'} \end{bmatrix} \begin{bmatrix} \dot{\mathbf{r}}_{O'} \\ \dot{\theta}_z \end{bmatrix} \\ &= \frac{1}{2} \begin{bmatrix} \dot{\mathbf{r}}_C^T & \dot{\theta}_z \end{bmatrix} \begin{bmatrix} m\mathbf{I}_2 & \mathbf{O}_{2 \times 1} \\ \mathbf{O}_{1 \times 2} & J_{z,C} \end{bmatrix} \begin{bmatrix} \dot{\mathbf{r}}_C \\ \dot{\theta}_z \end{bmatrix} \\ &= \frac{1}{2} m \dot{\mathbf{r}}_C^T \dot{\mathbf{r}}_C + \frac{1}{2} J_{z,C} \dot{\theta}_z^2 \end{aligned}$$

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坐标变换式	$\mathbf{r}'_k = \mathbf{A} \mathbf{l}'_k$
坐标变换阵	$\mathbf{A} = \begin{bmatrix} \cos(\theta_z) & -\sin(\theta_z) \\ \sin(\theta_z) & \cos(\theta_z) \end{bmatrix}$

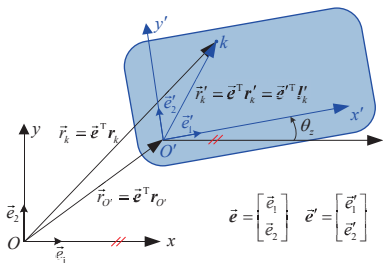
平面运动刚体元重力虚功可表示为

$$\sum_{k=1}^N \delta \mathbf{r}_k^T (m_k \mathbf{g}) = \delta \sum_{k=1}^N \mathbf{r}_k^T (m_k \mathbf{g}) = \delta \sum_{k=1}^N m_k \mathbf{g}^T \mathbf{r}_k =: \delta (-V_{\text{gravity}}),$$

式中

$$V_{\text{gravity}} = - \sum_{k=1}^N m_k \mathbf{g}^T \mathbf{r}_k$$

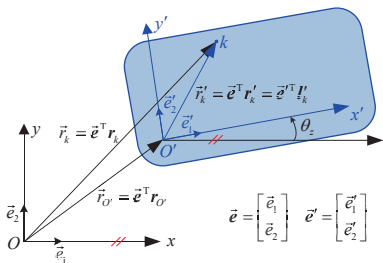
称为刚体元件的重力势能，可由系统的广义坐标确定。



坐标变换式	$\mathbf{r}'_k = \mathbf{A} \mathbf{l}'_k$
坐标变换阵	$\mathbf{A} = \begin{bmatrix} \cos(\theta_z) & -\sin(\theta_z) \\ \sin(\theta_z) & \cos(\theta_z) \end{bmatrix}$
重力势能	$V_{\text{gravity}} = -\sum_{k=1}^N m_k \mathbf{g}^T \mathbf{r}_k$
重力的虚功	$\delta W_{\text{gravity}} = \delta(-V_{\text{gravity}})$

刚体元件重力虚功的广义坐标表示形式:

$$\begin{aligned}
 \delta W_{\text{gravity}} &= \delta(-V_{\text{gravity}}(\mathbf{r}_1(\mathbf{q}), \mathbf{r}_2(\mathbf{q}), \dots, \mathbf{r}_N(\mathbf{q}))) \\
 &= \delta(-V_{\text{gravity}}(\mathbf{q})) \\
 &= \frac{\partial(-V_{\text{gravity}})}{\partial \mathbf{q}} \delta \mathbf{q} \\
 &= \delta \mathbf{q}^T \left(-\frac{\partial V_{\text{gravity}}}{\partial \mathbf{q}} \right)^T
 \end{aligned}$$



坐标变换式	$\mathbf{r}'_k = \mathbf{A} \mathbf{l}'_k$
坐标变换阵	$\mathbf{A} = \begin{bmatrix} \cos(\theta_z) & -\sin(\theta_z) \\ \sin(\theta_z) & \cos(\theta_z) \end{bmatrix}$
重力势能	$V_{\text{gravity}} = -\sum_{k=1}^N m_k \mathbf{g}^T \mathbf{r}_k$
重力的虚功	$\delta W_{\text{gravity}} = \delta(-V_{\text{gravity}})$

刚体元件的重力势能可进一步化简为如下形式

$$\begin{aligned}
 V_{\text{gravity}} &= -\mathbf{g}^T \sum_{k=1}^N m_k \mathbf{r}_k = -\mathbf{g}^T m \mathbf{r}_C = -m \mathbf{g}^T \mathbf{r}_C \\
 &= -\mathbf{g}^T \sum_{k=1}^N m_k (\mathbf{r}_{O'} + \mathbf{A} \mathbf{l}'_k) \\
 &= -\mathbf{g}^T (m \mathbf{r}_{O'} + m \mathbf{A} \mathbf{l}'_C) \\
 &= -m \mathbf{g}^T (\mathbf{r}_{O'} + \mathbf{A} \mathbf{l}'_C)
 \end{aligned}$$

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- ④ 弹性恢复力对应的广义力

线性弹簧弹性力的虚功

设微元 i 与微元 j 之间连接一个拉压刚度为常数 k 、原长为 l 的弹簧，则弹簧弹性恢复力的虚功可表示为

$$\delta W_{\text{spring}} = \delta \mathbf{r}_i^T \mathbf{f}_{i,j} + \delta \mathbf{r}_j^T \mathbf{f}_{j,i},$$

式中

$$\mathbf{f}_{i,j} = k (|\mathbf{r}_j - \mathbf{r}_i| - l) \mathbf{n}_{i,j}, \quad \mathbf{n}_{i,j} = \frac{\mathbf{r}_j - \mathbf{r}_i}{|\mathbf{r}_j - \mathbf{r}_i|},$$

$$\mathbf{f}_{j,i} = -\mathbf{f}_{i,j}.$$

可证

$$\delta W_{\text{spring}} = \delta \left[-\frac{1}{2} k (|\mathbf{r}_j - \mathbf{r}_i| - l)^2 \right].$$

线性弹簧弹性势能

线性弹簧弹性恢复力的虚功

$$\begin{aligned}\delta W_{\text{spring}} &= \delta \left[-\frac{1}{2}k (|\mathbf{r}_j - \mathbf{r}_i| - l)^2 \right] =: \delta (-V_{\text{spring}}(\mathbf{r}_i(\mathbf{q}), \mathbf{r}_j(\mathbf{q}))) \\ &= \delta (-V_{\text{spring}}(\mathbf{q})) \\ &= \frac{\partial (-V_{\text{spring}})}{\partial \mathbf{q}} \delta \mathbf{q} \\ &= \delta \mathbf{q}^T \left(-\frac{\partial V_{\text{spring}}}{\partial \mathbf{q}} \right)^T,\end{aligned}$$

式中

$$V_{\text{spring}} = \frac{1}{2}k (|\mathbf{r}_j - \mathbf{r}_i| - l)^2$$

为弹簧的弹性势能.

第二类拉格朗日方程

拉格朗日方程的一般形式

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{\mathbf{q}}} \right)^T - \left(\frac{\partial T}{\partial \mathbf{q}} \right)^T + \mathbf{c}_q^T \boldsymbol{\lambda} = \mathbf{Q}.$$

当系统广义坐标 $\mathbf{q}$ 完全独立，且系统受到的所有外力和内力都为有势力时，拉格朗日方程可转化为

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{\mathbf{q}}} \right)^T - \left(\frac{\partial T}{\partial \mathbf{q}} \right)^T = - \left(\frac{\partial V}{\partial \mathbf{q}} \right)^T.$$

上式可进一步简记为

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{\mathbf{q}}} \right) - \frac{\partial T}{\partial \mathbf{q}} = - \frac{\partial V}{\partial \mathbf{q}}.$$

第二类拉格朗日方程

当系统广义坐标 $\mathbf{q}$ 完全独立，且系统受到的所有外力和内力都为有势力时，拉格朗日方程可表示为

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{\mathbf{q}}} \right) - \frac{\partial T}{\partial \mathbf{q}} = - \frac{\partial V}{\partial \mathbf{q}}.$$

移项可得

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{\mathbf{q}}} \right) - \frac{\partial (T - V)}{\partial \mathbf{q}} = \mathbf{0}.$$

考虑到系统势能 V 的表达式仅与系统广义坐标 $\mathbf{q}$ 相关，与系统的广义速度 $\dot{\mathbf{q}}$ 无关，上式可进一步改写为

$$\frac{d}{dt} \left[\frac{\partial (T - V)}{\partial \dot{\mathbf{q}}} \right] - \frac{\partial (T - V)}{\partial \mathbf{q}} = \mathbf{0}.$$

第二类拉格朗日方程

当系统广义坐标 $\mathbf{q}$ 完全独立，且系统受到的所有外力和内力都为有势力时，拉格朗日方程可表示为

$$\frac{d}{dt} \left[\frac{\partial(T - V)}{\partial \dot{\mathbf{q}}} \right] - \frac{\partial(T - V)}{\partial \mathbf{q}} = \mathbf{0}.$$

通过引入拉格朗日函数

$$L(\mathbf{q}, \dot{\mathbf{q}}) = T(\mathbf{q}, \dot{\mathbf{q}}) - V(\mathbf{q}),$$

上述拉格朗日方程可进一步表示为如下第二类拉格朗日方程的形式

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\mathbf{q}}} \right) - \frac{\partial L}{\partial \mathbf{q}} = \mathbf{0}.$$